

A pedestrian detection algorithm based on deep learning

Jiangkun Lu

School of Computer Engineering, Chongqing College of
Humanities, Science & Technology, Chongqing, China,
851898930@qq.com, Corresponding author

Hongyang Chen

School of Computer Engineering, Chongqing College of
Humanities, Science & Technology, Chongqing, China
1060344275@qq.com

ABSTRACT

In order to solve the problem of crowd counting and crowd density statistics in the security intelligent video surveillance system, this paper adopts the method based on deep learning to optimize the algorithm. This method mainly uses the VGG16 backbone network with 1*1 and 3*3 small convolution Classification and feature extraction of people information in a crowd. In order to reduce the sharp reduction in the number of positive samples after increasing the threshold, and to avoid the situation where using different thresholds during training and testing will cause the performance of the detector to degrade, this paper draws on the cascade structure of the Cascade R-CNN network for input video. The frame images are analyzed and processed, and different IoU thresholds are set at different stages to obtain enough positive samples to reduce over-fitting, and use the multi-task loss function and the Hadamard product to obtain the pedestrian detection network, and output the final number of people. The improved pedestrian counting algorithm in this paper is tested in the public dataset WorldExpo'10 Crowd Counting, and compared with other algorithms to verify the feasibility and effectiveness of this method.

CCS CONCEPTS

• **Theory of computation** → Design and analysis of algorithms.

KEYWORDS

Dense Crowd Counting, Pedestrian Detection, Deep Learning

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1 INTRODUCTION

The research on pedestrian detection methods only studied the effective feature extraction and classification algorithms of pedestrians. With the introduction of a special training library for pedestrian detection, the accuracy of pedestrian detection algorithms has also been improved, and the increase in detection speed has led to the rapid development of pedestrian detection. After that, the neural

network density estimation algorithm based on deep learning was proposed, and the accuracy and efficiency of the results of crowd counting and crowd density estimation became higher and higher [1]. However, because pedestrians are easily affected by clothing, mutual occlusion, posture, and angle, the existing methods are still insufficient to accurately detect the number and density of crowds. Pedestrian detection is still a hot spot and difficulty in the current artificial intelligence field.

In 1985, the classical theory of crowd density was proposed by Fruin [2] et al.: When the density of the crowd exceeds a threshold (0.15m²/person), the possibility of gathering crowds out of control will greatly increase, which will lead to serious problems. Public safety accidents, crisis to social security. This theory specifies the correlation between crowd density and the occurrence of abnormal accidents, and also lays the foundation for the role of crowd density estimation in the detection of crowd abnormal events.

For the occlusion problem in images, researchers have been continuously improving the optimization prediction method, but most of the researchers just return the extracted effective pedestrian information to the global feature and the global number of people, thus ignoring the spatial information. But Lempitsky and Zisserman [3] et al. proposed a new supervised learning framework for the visual object counting task, avoiding the difficult task of learning to detect and localize individual object instances.

Object counting methods based on density estimation are more suitable for large-scale population statistics, have lower complexity, have better results in the case of small and dense objects in the image, and can basically reach the rate of real-time prediction. It is the most commonly used pedestrian counting system framework. However, due to various reasons such as different camera viewing angles and light differences, the shape of the same target object under different image color temperatures is very different, so the cross-scene counting performance of the model has great instability and generalization weaker. Although the density estimation method has better counting performance than other models, there are still various problems in practical scenarios, and there is still a lot of room for improvement.

Compared with traditional methods, the target counting method based on deep learning does not require manual selection of features, and allows the network to perform end-to-end learning through training, which has excellent performance and is also the focus of current research in the field. At present, the target detection methods based on deep learning are mainly divided into two categories: one stage and two stage [4]. The former is to directly convert the problem of target frame positioning into a regression problem. The latter is to first generate a series of sample candidate boxes by the algorithm, that is, to train the RPN (Region Proposal Network) network first, and then perform sample classification training through the convolutional neural network [5].

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2 THEORETICAL

Combined with the research background and research status of crowded crowd counting and crowd density algorithms, inspired by the successful application of convolutional neural networks in the field of artificial intelligence, this paper proposes a convolutional neural network based cascade structure for crowded crowd counting and crowd density. Analysis program. The network proposed in this paper is designed based on Cascade R-CNN and adopts a two-stage object detection framework to solve the pedestrian detection problem. The backbone network adopts VGG16, which is better introduced in the previous section. The structure and training process of the network will be described in detail below.

2.1 Pedestrian detection network based on deep learning

As mentioned above, the VGGNet[6] structure is optimized compared to other convolutional neural network structures such as AlexNet. In two aspects, one reduces the size of the convolution kernel of the convolutional layer, and the other increases the number of convolutional layers. better detection results. The feature extraction part of the target object adopts the backbone network structure of VGG16, and makes some appropriate modifications for our upcoming target population counting detection system.

Since it needs to be improved into a cascaded network structure, on the one hand, the pooling layer in the VGG16 network and the three FC (full connection layers) behind it are removed, and only the backbone network is used, and the parameters of the fully connected layer are large. It is not conducive to the cascade structure network after training. On the other hand, in the original VGG16 network, the four-stage pooling layer operation results in the input image of size $224 \times 224 \times 3$ being scaled to size $7 \times 7 \times 512$. In order to ensure the accuracy and operation speed of the subsequent cascaded network detection output, the final feature size should be increased.

Therefore, after careful thinking, change the pooling layer, reduce one pooling layer, and make the final feature map size smaller. Scale down to 1/8 of the original input size. The network backbone structure adopts smaller 1×1 convolution kernel and 3×3 convolution kernel. The 1×1 convolution kernel can preferentially reduce the dimension, increase the ability of nonlinear expression, and make the image segmentation reliable. At the same time, using 3×3 convolution kernels to superimpose multiple convolution kernels can effectively increase the spatial receptive field and reduce invalid parameters. However, the initial VGG16 network model has huge parameters and needs to consume huge storage space, and the speed of training convergence is relatively slow, which is prone to overfitting and other phenomena. Therefore, the fully connected layer is reduced in the improved network and the Cascade structure is added. Theoretical analysis shows that this structure can effectively reduce the model parameters, increase the diversity of positive samples, increase the training convergence speed and reduce the phenomenon of overfitting according to the cascade structure. The comparison shows that the feature extraction structure of the first part of the improved VGG16 network is consistent with the original VGG16 network structure. Only the number of pooling layers is changed, and the pedestrian image segmentation result and

the pedestrian image regression box result are predicted through the branch convolution layer, and then continue to be input into the following cascade structure, and finally the pedestrian density prediction result can be obtained.

2.2 Pedestrian detection algorithm based on Cascade-RCNN network structure

Cascade-RCNN consists of a series of detectors with different IoU thresholds. The entire network is delivered stage by stage with progressive training, the network at the current stage can provide better prediction boxes for the next stage of network training. The process of network processing and processing the picture to obtain the specific number of people is as follows. The input image is first processed through the VGG16 network conv1 block, conv2 block, and conv3 block. The conv4 block receives the features obtained from the first three blocks of VGG16, and some of the features are passed back to the conv5 block along the backbone network. The reverse connection module is fused with the high-level semantic information of the backbone network to obtain the red convolution depth feature under conv4. Part of the convolutional depth feature is transmitted to the instance semantic segmentation network branch for instance segmentation, reducing the number of candidate frames through the instance semantic segmentation information, and performing bias regression of the detection frame at the same time to enhance the accuracy of positioning; another part of the convolutional depth feature is processed by the inception module. The features processed by the inception module are fused with the remaining detection frame information after the instance segmentation network branch processing to perform pedestrian recognition. The left detection frame information is the position of the pedestrian in the image, and the pedestrian can be detected. The processing of conv5, conv6, and conv7 is similar to that of conv4. Since the feature maps of conv4, conv5, conv6, and conv7 have different sizes, the receptive fields represented by the feature maps are also different, which can solve the multi-scale problem in pedestrian detection and better detect pedestrians in the scene.

2.3 Multi-task loss function

The activation function of the neural convolutional network in this paper adopts the improved ReLU function Exponential Linear Units function. Aiming at the singularity defect of ReLU function, ELU function is proposed. Compared with the ReLU function, when the input is negative, the output has a certain anti-interference ability. This eliminates the problem of ReLU gradient saturation exponential operations. The ELU function formula is as follows:

$$f(x) = \begin{cases} x, & x > 0 \\ \alpha(e^x - 1), & x < 0 \end{cases} \quad (1)$$

When pedestrian detection handles multiple tasks, multiple tasks will bring multiple losses, and the definition and selection of the loss function is a critical decision. Therefore, choosing an appropriate loss function is also an important reason to ensure efficient training of the network detection model.

For pedestrian density bounding box regression, the candidate bounding box $b = (b_x, b_y, b_w, b_h)$ is regressed to the target bounding box g using the $f(x_i, b_i)$ regression, and the network calculates

the residual from the sample $\{g_i, b_i\}$ for learning, that is, the localization loss function is described as formula (2):

$$R_{loc} [f] = \sum_{i=1}^N L_{loc} (f(x_i, b_i), g_i) \quad (2)$$

The pedestrian foreground segmentation in the network model can be regarded as a classification task, and the loss function can be expressed as the following formula (3) through the minimization calculation:

$$R_{cls} [h] = \sum_{i=1}^N L_{cls} (h(x_i, y_i)) \quad (3)$$

Where (x_i, y_i) is the given sample training set, x_i is the sample, y_i is the sample class label, and h is the multi-dimensional estimation result, L_{cls} is the categorical cross-entropy loss function commonly used in R-CNN.

2.4 Post-processing of pedestrian detection

After the pedestrian counting network is trained, it is necessary to fuse the foreground segmentation results and the regression results for post-processing to output the final pedestrian detection results. This section will introduce the use of Hadamard product to post-process the pedestrian detection network to further generate the final pedestrian detection result. The specific process is to convert the result of the foreground segmentation prediction into the probability of the foreground effective information through the normalization function sigmoid. If the effective information exceeds 50%, it is considered that the event has occurred, and if the effective information is less than 50%, it is considered not to be occur. The prediction result of foreground segmentation is converted into probability and the Hadamard product is performed with the pedestrian density regression result, and then the pedestrian detection network outputs the final crowd number or crowd density.

In practical applications based on security surveillance video data, pedestrian detection systems have been tested many times and applied in different scenarios. Based on a large number of practical applications, it is found that the size of crowded crowd images is actually any size, not the size set in the training samples. Secondly, the crowded density distribution of crowds is not uniform. Due to the influence of angle and light occlusion, the size and saturation of the pixels occupied by individuals in different positions in the crowd are quite different, resulting in limited application places and inconvenience. Defects such as low stickiness and poor model generalization ability.

3 EXPERIMENTAL RESULTS AND ANALYSIS

Since the pedestrian counting scheme of the improved cascaded convolutional neural network proposed in this paper is mainly used to improve the problem of low pedestrian resolution and small pixels occupied by surveillance cameras in various scenes, the original intention of the subject is more to solve the problem of detection accuracy in different scenes. Combined with the overview of the pedestrian dataset described in Chapter 1, this topic selects the WorldExpo'10 Crowd Counting (hereinafter referred to as WE10) dataset, preprocesses the dataset and changes it to the same format.

The WE10 data set is a ten-minute video data sampling. The red pixel part of the data set is the marked pedestrian head feature. Because the number of people in different images in the WE10 data set will vary greatly, according to the number of people in a single image The crowd density is divided into three levels, the output is low density when the number of people is 0-20, the output is medium density when the number of people is 20-50, and the output is high density when the number of people is above 50.

3.1 Training experiment planning process

After obtaining the preprocessed sample image data by the method described above, how to train the experimental pedestrian detector is the next work. The closer the multi-task learning results of the density regression task and the foreground segmentation task obtained by the pedestrian detector are to the foreground segmentation map and the density regression map, the higher the accuracy is. The density regression map and foreground segmentation map obtained from the sample image are based on the original resolution of the image, so a pooling operation is required to make the size of the result output by the detector equal to the foreground segmentation map and the density regression map.

In this paper, a cascaded convolutional neural network is used for pedestrian counting detection. When a pedestrian detection model uses a fixed threshold to define positive samples and negative samples, then when the IoU size of the input detection frame is around this threshold, it means that the detection result of the detection model is better than the detection model effect when training with other thresholds. Be excellent. The network in this paper adopts the VGG backbone network to output segmentation and regression results. Through three cascade structures, three different thresholds are set to train to get the final result. The output of the previous pedestrian detection model in the network is used as the input of the next pedestrian detection model. This method is a typical stage to stage training method.

First, the candidate box regression results are generated based on the VGG16 network and the RPN network, and further input to the ROI pooling layer to map candidate boxes of different sizes. Since a single bounding box regressor in the VGG16 network cannot solve the problem of precise localization, bounding-box regression is used. The threshold of Bbox in Faster R-CNN is a fixed single threshold, but the specific value of IoU determines the accuracy of this network model, so the setting of a single threshold cannot well determine when it is the optimal solution. However, the convolutional neural network of the cascade structure ensures that there are enough positive samples in each stage by cascading and resampling the input hypothesis distribution of different stages, which solves the problem of overfitting described in Chapter 2. And by increasing the threshold to train the deep network, the accuracy of the network model is higher. Usually, the three-stage training thresholds are set as $a=0.5$, $b=0.6$, $c=0.7$, and the IoU threshold gradually increases.

3.2 Experimental results and analysis

We train the proposed pedestrian detection network model according to the training strategy described above. Select other tested pedestrian detection algorithms based on Faster R-CNN as a comparison. In this paper, the average precision Average Precision (AP)

Table 1: Comparison of detection accuracy of different networks

R-CNN	MOTA	AP	AP50	AP75
Cascade R-CNN	Resnet	42.8	62.1	46.3
Ours	VGG16	39.7	56.0	41.8

Table 2: Comparison of detection accuracy of different stage

Stage	AP50	AP60	AP70	AP80	AP90
Baseline	75.8	67.2	52.2	27.9	2.7
1	74.8	67.6	56.6	38.0	2.9
2	75.6	68.7	56.7	35.5	6.5
3	74.8	65.9	52.5	27.8	8.0
1+2	75.9	68.4	56.0	35.7	6.4
1+2+3	75.8	68.7	57.1	37.7	7.5

index is used to compare the methods. The AP calculation in this paper uses 11-point interpolation AP, that is, a group of 11 equally spaced recall levels [0, 0.1, 0.2,..., 1] on the average of precision value.

Comparing different network detection structures, as shown in table 1, it is found that the accuracy of pedestrian detection network based on VGG backbone network and cascade structure is far superior to one-stage series networks, such as YOLO, SSD and so on. However, it is lower than the detection accuracy of Cascade R-CNN based on residual network ResNet and cascade structure, but because it reduces the number of backbone network layers, the operation speed is faster than residual network ResNet and Cascade R-CNN based on cascade structure. Therefore, the feasibility of this method is verified, and the detection accuracy is improved compared with the general method, but it is not optimal.

As can be seen from Table 2, the accuracy of the pedestrian detector based on the convolutional neural network cascade structure in this paper increases with

The increase of the number of stages and the change of the threshold value have improved, showing that the cascaded multiple classifier indicators have improved, but the score of stage2 is significantly better than that of stage1 and stage3, and the training speed is slowed down after adding stage3, so for In practical application, the algorithm in this paper is better to use two stages. Although the effect is not very obvious, it proves the feasibility and research significance of the algorithm in this paper.

From the experimental results, we can see that the pedestrian detection method of the cascade structure has a certain improvement in accuracy, but it is not the optimal result, and further research is needed. However, the method in this paper has a large number of convolutional layers and a large number of iterative calculations, resulting in a slower running speed than the one-stage series of networks. Through the cascade structure analysis, the two-stage detection results in this method are better than the one-stage and three-stage detection results. Therefore, if the method in this paper adopts two stages, it can not only reduce the training running time,

but also increase the detection effect of the detector. Most of the current deep learning methods for estimating the number of people based on crowd density maps are not fast and cannot accurately estimate the number of people in the target scene in real time. In terms of running speed, the detection network proposed in this paper is slower than YOLO, SSD and other networks, and has a faster operation speed than Cascade R-CNN with higher accuracy. It is slightly optimized in general, which basically verifies the feasibility of the method in this paper. , and basically meet the practical application requirements of crowd density estimation.

4 CONCLUSION

In practical applications based on security surveillance video data, the pedestrian detection system has been tested and applied many times in different scenarios. In order to solve the problems of any size of crowded crowd images, uneven distribution of crowd density, and large differences in the size and saturation of pixels occupied by individuals in different positions in the crowd, this paper proposes a deep learning-based algorithm. The pedestrian detection algorithm adopts the VGG16 backbone network with 1*1 and 3*3 small convolution kernels to classify and extract features of different volumes of personnel information in the crowd. In order to reduce the sharp reduction in the number of positive samples after raising the threshold, and to avoid the situation that using different thresholds during training and testing will cause the performance of the detector to degrade, a cascade structure is designed, and different thresholds are set at different stages to obtain sufficient many positive samples. In the training process, the parameter hard sharing method is used, and the weighted summation result of the two loss functions is used, which increases the training time to a certain extent, but provides more global information, which enhances the generalization performance of the model and detection accuracy.

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